

1.

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GNU nano 8.7.1 Practical5b.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int fd[2];

    pipe(fd);

    if(fork() == 0)
    {
        dup2(fd[1], STDOUT_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("ls", "ls", "-l", NULL);
    }

    if(fork() == 0)
    {
        dup2(fd[0], STDIN_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("grep", "grep", ".c", NULL);
    }

    close(fd[0]);
    close(fd[1]);

    wait(NULL);
    wait(NULL);

    return 0;
}
```

OUTPUT :

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abhis@MIKHAIL-37:~$ nano Practical5b.c
abhis@MIKHAIL-37:~$ gcc Practical5b.c -o Practical5b
abhis@MIKHAIL-37:~$ ./Practical5b
-rw-r--r-- 1 abhis abhis  402 Aug 28 03:24 2.c
-rwxr-xr-x 1 abhis abhis 16216 Oct  1 09:18 Practical5b
-rw-r--r-- 1 abhis abhis  501 Oct  1 09:18 Practical5b.c
-rwxr-xr-x 1 abhis abhis 21392 Oct  1 03:06 append_stderr
-rw-r--r-- 1 abhis abhis  3559 Oct  1 03:12 append_stderr.c
-rwxr-xr-x 1 abhis abhis 22280 Oct  1 03:47 background_jobs
-rw-r--r-- 1 abhis abhis  4933 Oct  1 04:11 background_jobs.c
-rw-r--r-- 1 abhis abhis  182 Sep 18 04:01 bh.c
-rwxr-xr-x 1 abhis abhis 21248 Sep 29 09:24 cd_dispatch
-rw-r--r-- 1 abhis abhis  4005 Sep 29 09:40 cd_dispatch.c
-rw-r--r-- 1 abhis abhis  308 Oct  1 03:32 combined.log
-rwxr-xr-x 1 abhis abhis 22392 Oct  1 03:32 combined_complex
-rw-r--r-- 1 abhis abhis  6036 Oct  1 03:31 combined_complex.c
-rwxr-xr-x 1 abhis abhis 16040 Sep 29 03:14 cow_demo
-rw-r--r-- 1 abhis abhis  288 Sep 29 03:15 cow_demo.c
-rw-r--r-- 1 abhis abhis  422 Sep 18 03:45 er.c
-rw-r--r-- 1 abhis abhis  121 Oct  1 03:07 error.log
-rw-r--r-- 1 abhis abhis    0 Oct  1 03:20 error_summary.txt
-rwxr-xr-x 1 abhis abhis 21344 Sep 29 07:37 executor
-rw-r--r-- 1 abhis abhis  2702 Sep 29 07:43 executor.c
-rw-r--r-- 1 abhis abhis  4619 Sep 29 09:53 export_state.c
-rwxr-xr-x 1 abhis abhis 16208 Oct  1 09:10 fork_demo
-rw-r--r-- 1 abhis abhis  403 Oct  1 09:12 fork_demo.c
-rw-r--r-- 1 abhis abhis  4223 Sep 29 10:02 history_pipeline.c
-rw-r--r-- 1 abhis abhis  1129 Sep 16 06:48 history_shell.c
-rwxr-xr-x 1 abhis abhis 22072 Oct  1 05:32 integrated_shell
-rw-r--r-- 1 abhis abhis  3685 Oct  1 05:37 integrated_shell.c
-rwxr-xr-x 1 abhis abhis 21408 Sep 29 10:14 io_redirection
-rw-r--r-- 1 abhis abhis  3548 Sep 29 10:18 io_redirection.c
-rwxr-xr-x 1 abhis abhis 23752 Oct  1 04:45 job_listing_fg
-rw-r--r-- 1 abhis abhis  6471 Oct  1 04:51 job_listing_fg.c
-rwxr-xr-x 1 abhis abhis 25088 Oct  1 05:23 job_suspension_pgid
-rw-r--r-- 1 abhis abhis  6516 Oct  1 05:29 job_suspension_pgid.c
-rw-r--r-- 1 abhis abhis  382 Sep 18 03:57 ll.c
-rw-r--r-- 1 abhis abhis   29 Oct  1 03:06 log.txt
-rwxr-xr-x 1 abhis abhis 21176 Oct  1 05:41 memory_and_test_shell
-rw-r--r-- 1 abhis abhis  2520 Oct  1 05:48 memory_and_test_shell.c
-rw-r--r-- 1 abhis abhis  569 Sep 29 03:03 memory_layout.c

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-rw-r--r-- 1 abhis abhis 1134 Sep 29 03:57 memory_test.c
-rw-r--r-- 1 abhis abhis 10261 Oct 1 03:32 merged.log
-rwxr-xr-x 1 abhis abhis 16128 Oct 1 09:14 multi_fork
-rw-r--r-- 1 abhis abhis 359 Oct 1 09:14 multi_fork.c
-rw-r--r-- 1 abhis abhis 3353 Sep 29 10:11 multi_pipe.c
-rw-r--r-- 1 abhis abhis 5 Oct 1 03:19 output_users.txt
-rw-r--r-- 1 abhis abhis 5218 Sep 29 04:08 parser_test.c
-rw-r--r-- 1 abhis abhis 3365 Sep 29 09:05 path_waitpid.c
-rw-r--r-- 1 abhis abhis 140 Sep 18 03:28 pgm.c
-rwxr-xr-x 1 abhis abhis 23856 Oct 1 08:57 pipeline_perf_shell
-rw-r--r-- 1 abhis abhis 4787 Oct 1 09:04 pipeline_perf_shell.c
-rwxr-xr-x 1 abhis abhis 16248 Aug 24 05:27 process
-rw-r--r-- 1 abhis abhis 667 Aug 24 05:27 process.c
-rw-r--r-- 1 abhis abhis 235 Sep 29 03:10 program.c
-rw-r--r-- 1 abhis abhis 3225 Sep 16 07:59 quote_parser.c
-rw-r--r-- 1 abhis abhis 3722 Sep 29 04:17 quotes_parser.c
-rwxr-xr-x 1 abhis abhis 1451 Oct 1 05:50 run_tests.sh
-rw-r--r-- 1 abhis abhis 267 Oct 1 05:33 shell_errors.log
-rw-r--r-- 1 abhis abhis 3394 Sep 29 03:41 shell_history.c
-rwxr-xr-x 1 abhis abhis 25272 Oct 1 04:59 signal_bg_recovery
-rw-r--r-- 1 abhis abhis 6623 Oct 1 05:06 signal_bg_recovery.c
-rwxr-xr-x 1 abhis abhis 23024 Oct 1 05:11 signal_handler
-rw-r--r-- 1 abhis abhis 3028 Oct 1 05:14 signal_handler.c
-rw-r--r-- 1 abhis abhis 396 Aug 24 05:57 skill2.c
-rw-r--r-- 1 abhis abhis 60 Oct 1 03:07 stderr.log
-rw-r--r-- 1 abhis abhis 10196 Oct 1 03:07 stdout.log
-rw-r--r-- 1 abhis abhis 289 Oct 1 05:42 test_report.txt
-rw-r--r-- 1 abhis abhis 1718 Sep 16 07:02 tokenizer.c
-rwxr-xr-x 1 abhis abhis 21976 Sep 29 09:13 vars_dispatch
-rw-r--r-- 1 abhis abhis 4348 Sep 29 09:19 vars_dispatch.c
-rw-r--r-- 1 abhis abhis 2 Sep 29 10:15 word_count.txt
-rw-r--r-- 1 abhis abhis 411 Sep 18 03:40 ww.c
-rw-r--r-- 1 abhis abhis 575 Sep 18 03:56 zx.c

```

2.

```

GNU nano 8.7.1 signal_demo.c
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handler(int sig)
{
    if(sig == SIGINT)
        printf("\nSIGINT Received\n");

    if(sig == SIGTERM)
        printf("\nSIGTERM Received\n");

    if(sig == SIGUSR1)
        printf("\nSIGUSR1 Received\n");
}

int main()
{
    signal(SIGINT, handler);
    signal(SIGTERM, handler);
    signal(SIGUSR1, handler);

    while(1)
    {
        printf("Running...\n");
        sleep(3);
    }

    return 0;
}

```

OUTPUT :

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abhis@MIKHAIL-37:~$ nano signal_demo.c
abhis@MIKHAIL-37:~$ gcc signal_demo.c -o signal_demo
abhis@MIKHAIL-37:~$ ./signal_demo
Running...
Running...
Running...
Running...
Running...
Running...
^C
SIGINT Received

```